

Laird Scabar

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EDUCATION

University of British Columbia

Bachelor of Applied Science — Mechanical Engineering, (Guaranteed Mechatronics Entry)

Vancouver, BC

GPA: 4.30/4.33

SKILLS

Design & Analysis:

CAD (SolidWorks, CATIA), Catia, FEA (Ansys), DFM/DFA, PCB (Altium)

Manufacturing & Prototyping:

3D Printing (FDM), Waterjet, CNC, Welding

Programming & Data Analysis:

MATLAB, C, C++, Python, ESP32, VR/XR

EXPERIENCE

Mechanical Engineering Intern – Thermal/HVAC

May 2026 – Sept 2026

VETS Group

Burnaby, BC

- Modelled custom sheet metal parts and assemblies in **SolidWorks** and **Advanced Steel** to **SMACNA** and **ASHRAE** standards, applying **DFM/DFA** so parts reached the shop ready to cut, brake, and seam.
- Released 80+ fabrication and general arrangement drawing packages to shop production, routing duct systems in **Revit** and coordinating with clients and the shop on **constructability** before release.
- Designed custom **structural duct supports** with a structural engineer and one-off snow hoods on request, confirming routing against existing and planned construction during **site visits**.
- Repaired existing company **SolidWorks** models to restore **parametric control** and rebuild reliability, making them reusable on later jobs instead of remodelled from scratch.
- Cut drawing setup time per assembly by developing a **Revit Dynamo** script that auto-generates spools, sheets, and **BOMs**, plus a **Revit MCP** tool for model-wide duct spec changes.

SUBC - UBC's Submarine Design Team

Sept 2025 – Present

Drive Train Sub-team Member

Vancouver, BC

- Designed and validated drivetrain and 2:1 **straight-bevel gearbox** components for a human-powered submarine under tight packaging, alignment, and **GD&T** tolerance constraints for a 1.8–2.1 kN design load.
- Reduced side plate mass 37% (1.89 lb → 1.19 lb) through **FEA**-guided material removal while maintaining required **safety factors** under peak torque, then **CNC**-machined and installed the part in the sub's gearbox.
- Designed a modified 0.500 in to 0.375 in step-down collar in **SolidWorks** for the driving pinion through **CAD** iteration and **fit checks**, eliminating slippage and restoring gearbox alignment.

PROJECTS

VR-Controlled Teleoperated Robotic Arm | *Personal Project*

July 2026 – Present

- Designed and built a **5-DOF** arm from scratch in **SolidWorks**, carrying joint loads on **bearings** rather than servo output shafts and mirroring two shoulder servos across a single pivot to share torque.
- Reduced part count and build time across 3 design iterations by tuning print orientation, **press-fit tolerances**, and fastener access (**DFM/DFA**), so the arm assembles from **FDM** parts and off-the-shelf hardware.
- Packaged 7 servos, an **ESP32**, and an external power supply into the printed structure on my Bambu Labs A1.
- Drove the arm in real time from a VR headset over a **WebXR** app solving closed-form **inverse kinematics (IK)** at 50 Hz, with **C++** firmware enforcing per-joint velocity and acceleration limits over serial.

Autonomous Robotic Claw | *C++, SolidWorks (Sheet Metal), Rapid Prototyping*

January 2026

- Led the design and fabrication of a **sheet metal** robotic claw in **SolidWorks**, detecting and grasping objects with an **ultrasonic sensor** and a servo-driven gripper.
- Programmed an **Arduino** microcontroller in **C++** with distance-based actuation logic for autonomous pickup.
- Integrated mechanical, electrical, and software subsystems, troubleshooting **sensor noise** and **servo calibration** to improve detection reliability to over 95% in testing.

Self-Balancing Robot | *C++, Arduino, MPU-6050, Control Systems*

December 2025

- Built a two-wheel self-balancing robot with an **MPU-6050 IMU** and **sensor fusion** for real-time tilt estimation.
- Implemented a **PID control** algorithm to maintain dynamic stability under external disturbance.
- Developed **closed-loop** motor control with an **L298N** driver to regulate dual DC motors from feedback signals.

ADDITIONAL EXPERIENCE

Film & Television Actor

2015–2025

- Performed in professional productions, developing strong communication, adaptability, and teamwork skills.
- Worked with companies including Netflix, Disney, CBC, Pfizer, and more.